

Coordinate Geometry in the (x, y) Plane

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2014 January Q11 C34

11. The curve C has parametric equations

$$x = 10 \cos 2t, \quad y = 6 \sin t, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

The point A with coordinates $(5, 3)$ lies on C .

(a) Find the value of t at the point A .

(1)

(b) Show that an equation of the normal to C at A is

$$3y = 10x - 41$$

(6)

The normal to C at A cuts C again at the point B .

(c) Find the exact coordinates of B .

(8)

9.

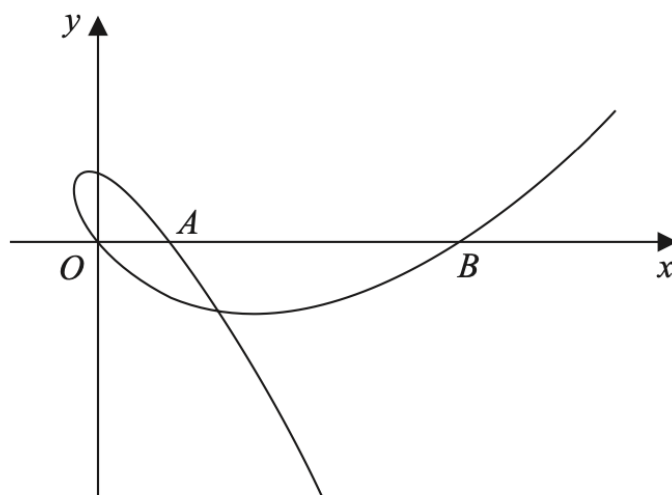


Figure 3

Figure 3 shows a sketch of part of the curve with parametric equations

$$x = t^2 + 2t, \quad y = t^3 - 9t, \quad t \in \mathbb{R}$$

The curve cuts the x -axis at the origin and at the points A and B as shown in Figure 3.

(a) Find the coordinates of point A and show that point B has coordinates $(15, 0)$. (3)

(b) Show that the equation of the tangent to the curve at B is $9x - 4y - 135 = 0$ (5)

The tangent to the curve at B cuts the curve again at the point X .

(c) Find the coordinates of X . (5)

(Solutions based entirely on graphical or numerical methods are not acceptable.)

11.

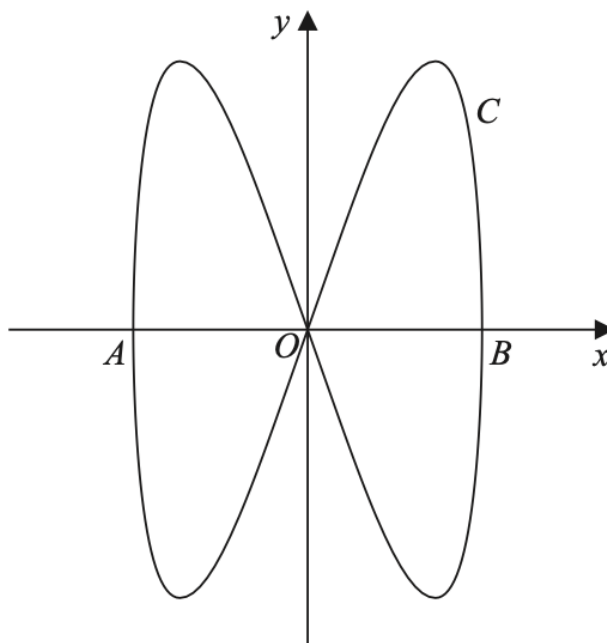


Figure 3

The curve C shown in Figure 3 has parametric equations

$$x = 3 \cos t, \quad y = 9 \sin 2t, \quad 0 \leq t \leq 2\pi$$

The curve C meets the x -axis at the origin and at the points A and B , as shown in Figure 3.

(a) Write down the coordinates of A and B . (2)

(b) Find the values of t at which the curve passes through the origin. (2)

(c) Find an expression for $\frac{dy}{dx}$ in terms of t , and hence find the gradient of the curve when $t = \frac{\pi}{6}$ (4)

(d) Show that the cartesian equation for the curve C can be written in the form

$$y^2 = ax^2(b - x^2)$$

where a and b are integers to be determined.

(4)

2018 June Q2 C34

2. A curve C has parametric equations

$$x = \frac{3}{2}t - 5, \quad y = 4 - \frac{6}{t} \quad t \neq 0$$

(a) Find the value of $\frac{dy}{dx}$ at $t = 3$, giving your answer as a fraction in its simplest form. (3)

(b) Show that a cartesian equation of C can be expressed in the form

$$y = \frac{ax + b}{x + 5} \quad x \neq k$$

where a , b and k are integers to be found.

(4)

12.

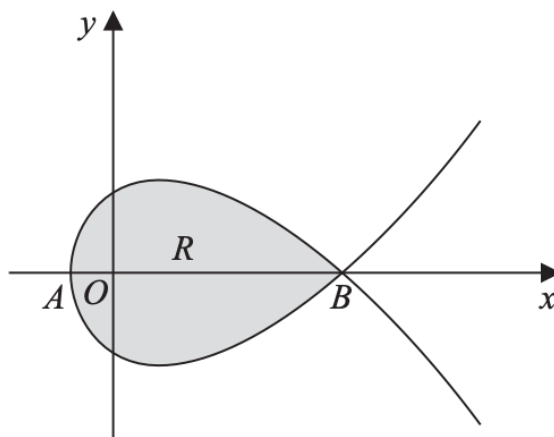


Figure 3

Figure 3 shows a sketch of part of the curve C with parametric equations

$$x = 7t^2 - 5, \quad y = t(9 - t^2), \quad t \in \mathbb{R}$$

- (a) Find an equation of the tangent to C at the point where $t = 1$

Write your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(5)

The curve C cuts the x -axis at the points A and B , as shown in Figure 3

- (b) (i) Find the x coordinate of the point A .

- (ii) Find the x coordinate of the point B .

(3)

The region R , shown shaded in Figure 3, is enclosed by the loop of the curve C .

- (c) Use integration to find the area of R .

(5)

2019 January Q8 C34

8. A curve has parametric equations

$$x = t^2 - t, \quad y = \frac{4t}{1-t} \quad t \neq 1$$

(a) Find $\frac{dy}{dx}$ in terms of t , giving your answer as a simplified fraction.

(4)

(b) Find an equation for the tangent to the curve at the point P where $t = -1$, giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(4)

The tangent to the curve at P cuts the curve at the point Q .

(c) Use algebra to find the coordinates of Q .

(5)

2019 June Q3 C34

3. A curve C has parametric equations

$$x = \sqrt{3} \tan \theta, \quad y = \sec^2 \theta, \quad 0 \leq \theta \leq \frac{\pi}{3}$$

The cartesian equation of C is

$$y = f(x), \quad 0 \leq x \leq k, \quad \text{where } k \text{ is a constant}$$

- (a) State the value of k . (1)
- (b) Find $f(x)$ in its simplest form. (2)
- (c) Hence, or otherwise, find the gradient of the curve at the point where $\theta = \frac{\pi}{6}$ (3)

4.

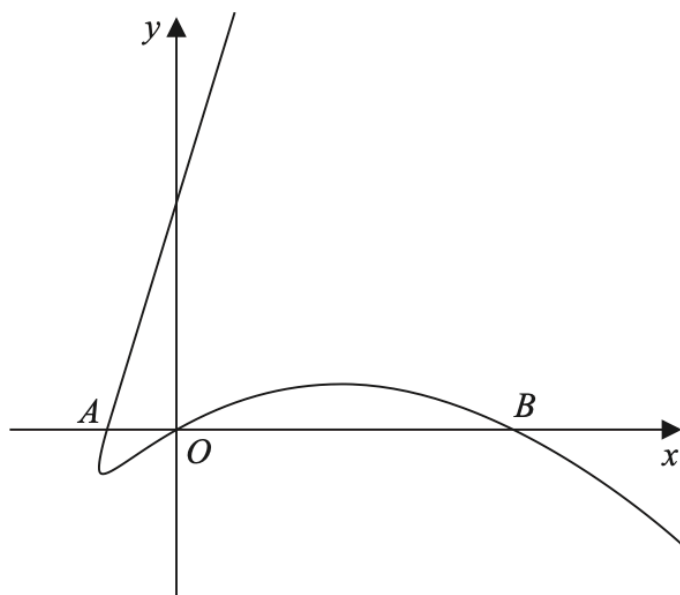


Figure 2

Figure 2 shows a sketch of part of the curve with parametric equations

$$x = 2t^2 - 6t, \quad y = t^3 - 4t, \quad t \in \mathbb{R}$$

The curve cuts the x -axis at the origin and at the points A and B , as shown in Figure 2.

(a) Find the coordinates of A and show that B has coordinates $(20, 0)$.

(3)

(b) Show that the equation of the tangent to the curve at B is

$$7y + 4x - 80 = 0$$

(5)

The tangent to the curve at B cuts the curve again at the point P .

(c) Find, using algebra, the x coordinate of P .

(4)

2021 January Q4 P4

4. The curve C is defined by the parametric equations

$$x = \frac{1}{t} + 2 \quad y = \frac{1 - 2t}{3 + t} \quad t > 0$$

- (a) Show that the equation of C can be written in the form $y = g(x)$ where g is the function

$$g(x) = \frac{ax + b}{cx + d} \quad x > k$$

where a, b, c, d and k are integers to be found.

(5)

- (b) Hence, or otherwise, state the range of g .

(2)

6.

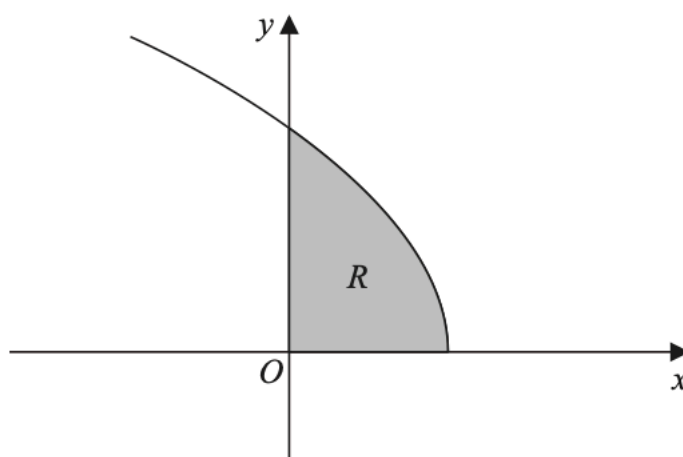


Figure 3

Figure 3 shows a sketch of the curve C with parametric equations

$$x = 2 \cos 2t \quad y = 4 \sin t \quad 0 \leq t \leq \frac{\pi}{2}$$

The region R , shown shaded in Figure 3, is bounded by the curve, the x -axis and the y -axis.

(a) (i) Show, making your working clear, that the area of $R = \int_0^{\frac{\pi}{4}} 32 \sin^2 t \cos t \, dt$

(ii) Hence find, by algebraic integration, the exact value of the area of R . (6)

(b) Show that all points on C satisfy $y = \sqrt{ax + b}$, where a and b are constants to be found. (3)

The curve C has equation $y = f(x)$ where f is the function

$$f(x) = \sqrt{ax + b} \quad -2 \leq x \leq 2$$

and a and b are the constants found in part (b).

(c) State the range of f . (1)

5.

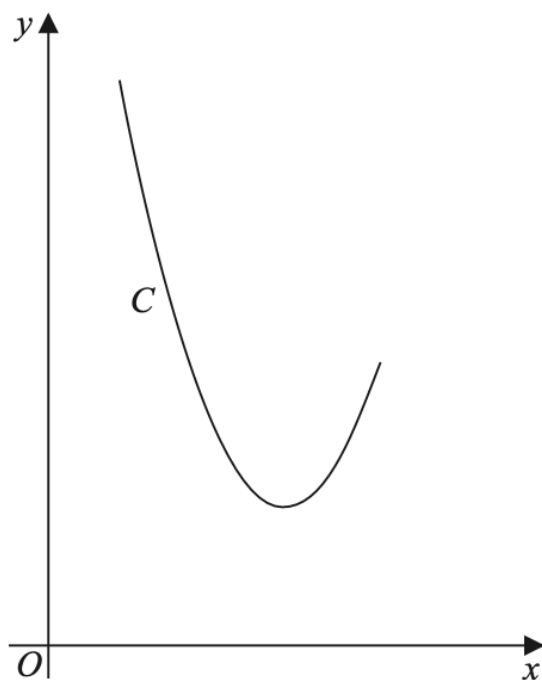


Figure 1

Figure 1 shows a sketch of the curve C with parametric equations

$$x = 5 + 2 \tan t \quad y = 8 \sec^2 t \quad -\frac{\pi}{3} \leq t \leq \frac{\pi}{4}$$

(a) Use parametric differentiation to find the gradient of C at $x = 3$

(4)

The curve C has equation $y = f(x)$, where f is a quadratic function.

(b) Find $f(x)$ in the form $a(x + b)^2 + c$, where a , b and c are constants to be found.

(3)

(c) Find the range of f .

(2)

2022 January Q3 P4

3. The curve C has parametric equations

$$x = 3 + 2 \sin t \qquad y = \frac{6}{7 + \cos 2t} \qquad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

(a) Show that C has Cartesian equation

$$y = \frac{12}{(7 - x)(1 + x)} \qquad p \leq x \leq q$$

where p and q are constants to be found.

(6)

(b) Hence, find a Cartesian equation for C in the form

$$y = \frac{a}{x + b} + \frac{c}{x + d} \qquad p \leq x \leq q$$

where a , b , c and d are constants.

(3)

2022 June Q7 P4

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has parametric equations

$$x = \sin t - 3 \cos^2 t \quad y = 3 \sin t + 2 \cos t \quad 0 \leq t \leq 5$$

- (a) Show that $\frac{dy}{dx} = 3$ where $t = \pi$ (4)

The point P lies on C where $t = \pi$

- (b) Find the equation of the tangent to the curve at P in the form $y = mx + c$ where m and c are constants to be found. (3)

Given that the tangent to the curve at P cuts C at the point Q

- (c) show that the value of t at point Q satisfies the equation

$$9 \cos^2 t + 2 \cos t - 7 = 0 \quad (2)$$

- (d) Hence find the exact value of the y coordinate of Q (3)

2022 October Q1 P4

1. A curve C has parametric equations

$$x = \frac{t}{t-3} \quad y = \frac{1}{t} + 2 \quad t \in \mathbb{R} \quad t > 3$$

Show that all points on C lie on the curve with Cartesian equation

$$y = \frac{ax-1}{bx}$$

where a and b are constants to be found.

(3)

6.

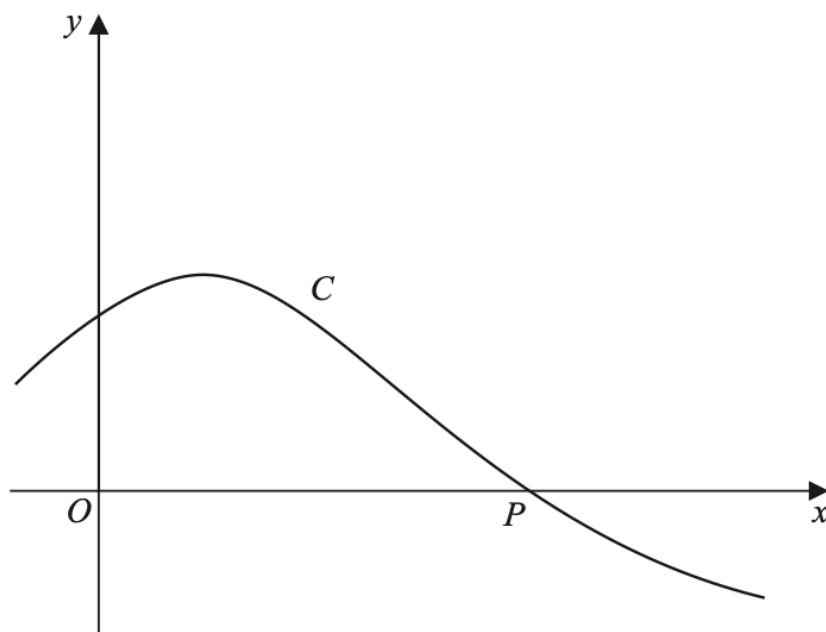


Figure 3

Figure 3 shows a sketch of the curve C with parametric equations

$$x = 1 + 3 \tan t \quad y = 2 \cos 2t \quad -\frac{\pi}{6} \leq t \leq \frac{\pi}{3}$$

The curve crosses the x -axis at point P , as shown in Figure 3.

- (a) Find the equation of the tangent to C at P , writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(5)

The curve C has equation $y = f(x)$, where f is a function with domain $\left[k, 1 + 3\sqrt{3} \right]$

- (b) Find the exact value of the constant k .

(1)

- (c) Find the range of f .

(2)

2023 January Q2 P4

2. A set of points $P(x, y)$ is defined by the parametric equations

$$x = \frac{t-1}{2t+1} \quad y = \frac{6}{2t+1} \quad t \neq -\frac{1}{2}$$

(a) Show that all points $P(x, y)$ lie on a straight line.

(4)

(b) Hence or otherwise, find the x coordinate of the point of intersection of this line and the line with equation $y = x + 12$

(2)

2023 January Q8 P4

8.

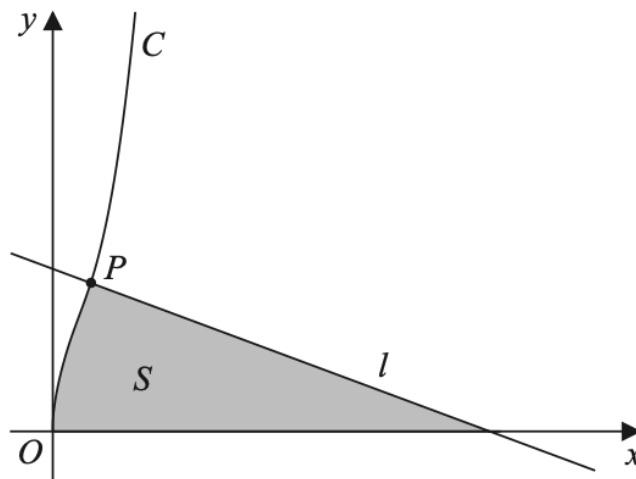


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A curve C has parametric equations

$$x = \sin^2 t \quad y = 2 \tan t \quad 0 \leq t < \frac{\pi}{2}$$

The point P with parameter $t = \frac{\pi}{4}$ lies on C .

The line l is the normal to C at P , as shown in Figure 3.

(a) Show, using calculus, that an equation for l is

$$8y + 2x = 17$$

(5)

The region S , shown shaded in Figure 3, is bounded by C , l and the x -axis.

(b) Find, using calculus, the exact area of S .

(6)

8.

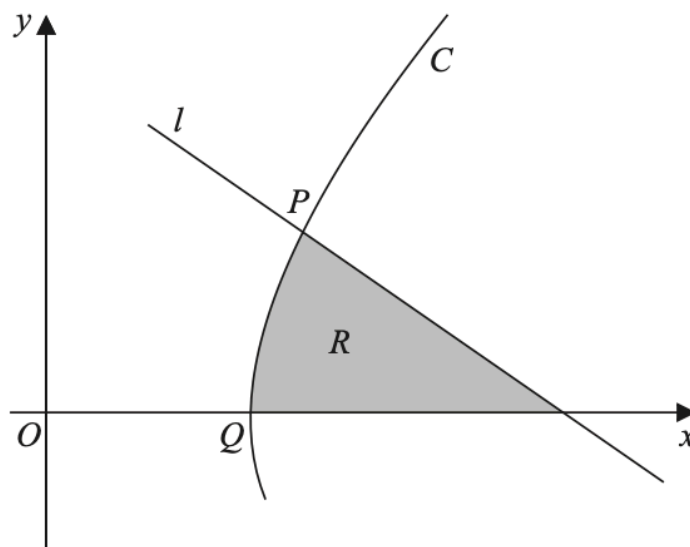


Figure 2

Figure 2 shows a sketch of part of the curve C with parametric equations

$$x = t + \frac{1}{t} \quad y = t - \frac{1}{t} \quad t > 0.7$$

The curve C intersects the x -axis at the point Q .

(a) Find the x coordinate of Q .

(1)

The line l is the normal to C at the point P as shown in Figure 2.

Given that $t = 2$ at P

(b) write down the coordinates of P

(1)

(c) Using calculus, show that an equation of l is

$$3x + 5y = 15$$

(3)

The region, R , shown shaded in Figure 2 is bounded by the curve C , the line l and the x -axis.

(d) Using algebraic integration, find the exact volume of the solid of revolution formed when the region R is rotated through 2π radians about the x -axis.

(7)

8.

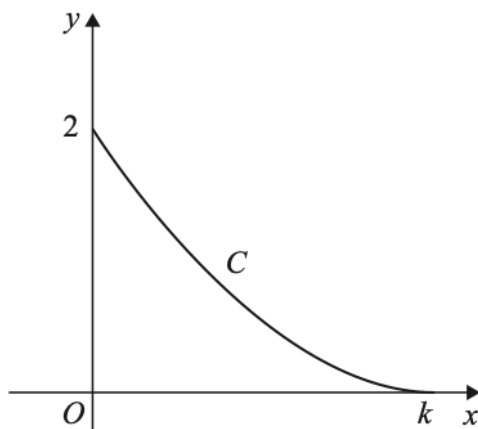


Figure 3

Figure 3 shows a sketch of the curve C with parametric equations

$$x = 6t - 3\sin 2t \quad y = 2\cos t \quad 0 \leq t \leq \frac{\pi}{2}$$

The curve meets the y -axis at 2 and the x -axis at k , where k is a constant.

(a) State the value of k .

(1)

(b) Use parametric differentiation to show that

$$\frac{dy}{dx} = \lambda \operatorname{cosec} t$$

where λ is a constant to be found.

(4)

The point P with parameter $t = \frac{\pi}{4}$ lies on C .

The tangent to C at the point P cuts the y -axis at the point N .

(c) Find the exact y coordinate of N , giving your answer in simplest form.

(3)

The region bounded by the curve, the x -axis and the y -axis is rotated through 2π radians about the x -axis to form a solid of revolution.

(d) (i) Show that the volume of this solid is given by

$$\int_0^\alpha \beta(1 - \cos 4t) dt$$

where α and β are constants to be found.

(ii) Hence, using algebraic integration, find the exact volume of this solid.

(6)

9.

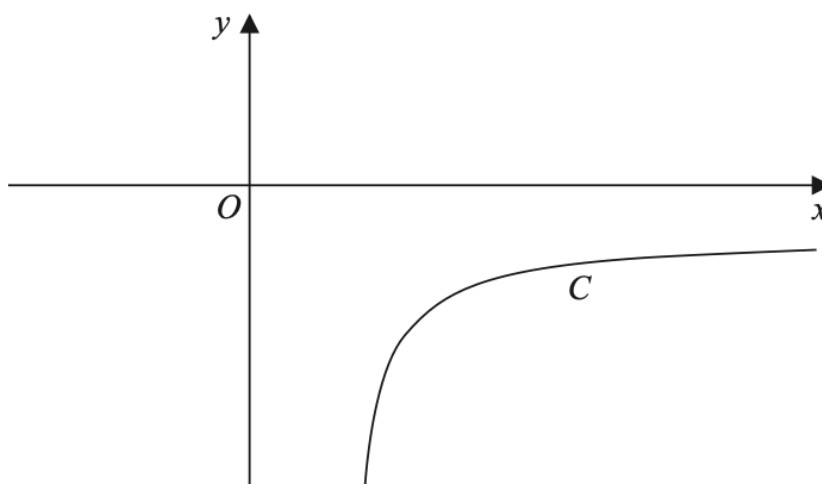


Figure 4

Figure 4 shows a sketch of the curve C with parametric equations

$$x = \sec t \quad y = \sqrt{3} \tan\left(t + \frac{\pi}{3}\right) \quad \frac{\pi}{6} < t < \frac{\pi}{2}$$

(a) Find $\frac{dy}{dx}$ in terms of t

(3)

(b) Find an equation for the tangent to C at the point where $t = \frac{\pi}{3}$

Give your answer in the form $y = mx + c$, where m and c are constants.

(4)

(c) Show that all points on C satisfy the equation

$$y = \frac{Ax^2 + B\sqrt{3x^2 - 3}}{4 - 3x^2}$$

where A and B are constants to be found.

(5)